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Liquidity Regulation and Banks: Theory and Evidence

Contents

1	Big picture	3
2	Data and measurement	3
2.1	LCR regulation and LCR banks	3
2.2	Liquidity ratio and exposure to regulation	3
2.3	Bank balance sheet data	4
2.4	Small-business lending and mortgage data	4
2.5	Spillover exposure of non-LCR banks	4
3	Empirical specifications	4
3.1	Direct balance-sheet effects on LCR banks	4
3.2	Lending effects on LCR banks	5
3.3	Spillover effects on non-LCR banks	5
4	Model	5
4.1	Intermediaries and balance sheets	5
4.2	Liquidity demand and liquidity premium	5
4.3	Laissez-faire equilibrium	6
4.4	First-best and Pigovian tax	6
4.5	LCR as implicit tax and costly mitigator	6
4.6	Welfare under the LCR	6
4.7	Uncertain liquidity demand and procyclicality	6
4.8	Committed liquidity facility	6
5	Identification and empirical design	7
5.1	What identifies the direct effect of LCR exposure	7
5.2	How the paper addresses loan-demand concerns	7
5.3	How the paper identifies spillovers	7
6	Tables and figures	7
6.1	Tables 1–3: summary statistics	7
6.2	Figures 1–3: aggregate trends and event-study evidence	8
6.3	Tables 4–5: HQLA and illiquid asset holdings	9
6.4	Tables 6–7: lending effects on LCR banks	10
6.5	Table 8: spillover to non-LCR banks	11
6.6	Figures 4–6: model equilibria and regulatory design	12
6.7	Tables 9–10: calibration and counterfactuals	12
7	Short summary	13
8	Conclusion	14
8.1	Core conclusion	14
8.2	Economic intuition	14

8.3	Takeaway	14
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1 Big picture

- **Question.** What are the effects of the liquidity coverage ratio (LCR) on banks, lending, risk migration, and welfare?
- **Empirical setting.** The paper studies U.S. post-crisis liquidity regulation. The LCR requires large banks to hold high-quality liquid assets (HQLA) at least as large as expected 30-day stressed net cash outflows.
- **Main empirical result.** LCR banks increased liquidity buffers after the rule, especially those whose pre-regulation liquidity ratios were further below the required level.
- **Crowding-out result.** More-exposed LCR banks reduced illiquid assets and lending. The lending decline is shown in CRA small-business lending and HMDA mortgage origination data.
- **Migration result.** Liquidity risks migrate to non-LCR banks. Non-LCR banks operating in markets with more LCR-bank exposure increase lending, while their liquidity ratios deteriorate.
- **Theory.** The model features liquidity transformation, fire-sale externalities, heterogeneous regulated and unregulated intermediaries, and endogenous liquidity premia.
- **Key theoretical mechanism.** The LCR liquidity buffer has a dual role: it is an implicit tax on liquidity transformation and also a costly mitigator of liquidity risk. Because the same quantity instrument plays both roles, it generally cannot achieve first-best lending and liquidity.
- **Distortion 1: crowding out.** If the mandated buffer is high enough to reduce short-term debt to the socially optimal level, it can force banks to hold too much low-return liquidity and lend too little.
- **Distortion 2: migration.** Regulated banks' extra HQLA demand raises the liquidity premium, encouraging unregulated intermediaries to expand liquidity transformation.
- **Distortion 3: procyclicality.** With uncertain liquidity demand, a fixed quantity mandate becomes more punitive when liquidity demand is high because the liquidity premium rises.
- **Policy result.** A central bank committed liquidity facility (CLF) adds a price-based mechanism to the quantity-based LCR and can improve welfare by reducing crowding out and migration.
- **Takeaway.** Liquidity regulation lowers liquidity risk at regulated banks, but its design matters. Quantity mandates can create credit, migration, and volatility distortions; a committed liquidity facility can mitigate them.

2 Data and measurement

2.1 LCR regulation and LCR banks

- The LCR was introduced after the financial crisis as a global liquidity standard.
- U.S. regulators implemented the LCR in 2013.
- The requirement applies primarily to large banking organizations.
- LCR banks must hold enough HQLA to cover expected net cash outflows over a 30-day stress period.

Interpretation: The LCR is a balance-sheet regulation: it changes both the asset side through HQLA holdings and the liability side through the treatment of unstable short-term funding.

2.2 Liquidity ratio and exposure to regulation

The core treatment intensity is the liquidity ratio gap:

$$\text{Gap}_i = \text{Required ratio}_i - \text{Pre-regulation ratio}_i. \quad (1)$$

- *Required ratio_i* is the required liquidity ratio under the LCR.
- *Pre-regulation ratio_i* is the bank’s liquidity ratio before LCR implementation.
- A larger gap means the bank is more exposed to the new regulation.

Interpretation: The gap is a continuous treatment measure. Banks with a larger gap need to make larger balance-sheet adjustments to comply.

2.3 Bank balance sheet data

- Bank balance-sheet data come from Call Reports and FR Y-9C.
- The key outcomes include liquidity coverage ratios, HQLA growth, illiquid asset growth, deposit growth, non-interest income, leverage, risk-weighted leverage, and exposure to other regulations.
- The baseline period is 2011Q1 to 2017Q4.

Interpretation: The bank data measure the direct response of regulated and unregulated banks to LCR implementation.

2.4 Small-business lending and mortgage data

- Small-business lending data come from the Community Reinvestment Act (CRA).
- The unit of observation is a bank–MSA–year.
- Mortgage origination data come from HMDA.
- Mortgage outcomes are split into conforming and non-conforming loans.

Interpretation: These loan data allow the authors to test whether mandated liquidity buffers crowd out credit provision.

2.5 Spillover exposure of non-LCR banks

For spillover tests, the paper defines the exposure of a local market to LCR banks:

$$LCR\text{-}bank\text{ share}_m = \frac{\sum_j LCRbank_j \times Loans_{m,j}}{\sum_j Loans_{m,j}}. \quad (2)$$

Interpretation: This variable captures whether a local lending market is dominated by regulated banks. It is used to test whether non-LCR banks expand in markets where LCR banks pull back.

3 Empirical specifications

3.1 Direct balance-sheet effects on LCR banks

The main bank-level regression is:

$$Y_{i,t} = \beta Post_t \times Gap_i + \gamma X_{i,t} + \tau_i + \tau_t + \epsilon_{i,t}. \quad (3)$$

- $Y_{i,t}$ is HQLA growth, illiquid asset growth, or another bank outcome.
- $Post_t$ indicates the post-LCR period.
- τ_i are bank fixed effects.
- τ_t are time fixed effects.

Interpretation: The coefficient β measures how much more strongly high-gap banks adjust after LCR implementation.

3.2 Lending effects on LCR banks

The local lending regression is:

$$Loan_{i,m,t} = \beta Post_t \times Gap_i + \gamma X_{i,m,t} + \tau_i + \tau_m + \epsilon_{i,m,t}. \quad (4)$$

Interpretation: Bank fixed effects absorb permanent bank differences. MSA fixed effects absorb local market characteristics. The coefficient tests whether exposed LCR banks reduce lending more after regulation.

3.3 Spillover effects on non-LCR banks

For non-LCR banks, the paper estimates:

$$Loan_{i,m,t} = \beta Post_t \times LCR\text{-}bank\ share_m + \gamma X_{i,m,t} + \tau_t + \tau_m + \epsilon_{i,m,t}. \quad (5)$$

The liquidity-ratio spillover regression is:

$$Liquidity_{i,t} = \beta Post_t \times Average\ LCR\text{-}bank\ share_i + \gamma X_{i,t} + \tau_i + \tau_t + \epsilon_{i,t}. \quad (6)$$

Interpretation: If liquidity regulation pushes lending and liquidity risk outside the regulated sector, non-LCR banks should increase lending and reduce liquidity in areas more exposed to LCR banks.

4 Model

4.1 Intermediaries and balance sheets

There are regulated and unregulated intermediaries. Regulated banks face the LCR; unregulated intermediaries do not. Banks choose short-term debt, liquid asset holdings, and lending.

The aggregate short-term debt, liquidity buffer, and lending are:

$$D = wd_1 + (1 - w)d_0, \quad (7)$$

$$L = wl_1 + (1 - w)l_0, \quad (8)$$

$$I = wi_1 + (1 - w)i_0, \quad (9)$$

where w is the fraction of regulated banks.

Interpretation: The model explicitly studies heterogeneous intermediaries and migration of activity between regulated and unregulated banks.

4.2 Liquidity demand and liquidity premium

The liquidity premium is determined by aggregate liquidity demand:

$$p = p(Q, \xi), \quad (10)$$

where Q is the stock of liquid assets and ξ is a liquidity-demand shock. Market clearing requires

$$D - L + G = Q. \quad (11)$$

Interpretation: The liquidity premium is endogenous. When regulated banks demand more HQLA, the premium rises and changes incentives for all intermediaries.

4.3 Laissez-faire equilibrium

In laissez-faire, banks do not internalize the fire-sale externality from excessive short-term debt. The privately optimal short-term debt satisfies a marginal private cost condition:

$$p = c'(d) + (1 - a)\mu\phi + ap. \quad (12)$$

Interpretation: The private equilibrium can feature too much short-term debt and too little liquidity because banks do not internalize systemic liquidity externalities.

4.4 First-best and Pigovian tax

Proposition 1. A Pigovian tax on short-term debt can implement the first-best equilibrium.

Interpretation: A direct tax on short-term debt targets the externality directly. The LCR is more complicated because its liquidity buffer also changes banks' asset allocation.

4.5 LCR as implicit tax and costly mitigator

The LCR requires regulated banks to hold liquid assets proportional to short-term debt:

$$l_1 \geq \rho d_1. \quad (13)$$

Proposition 2. The LCR imposes an implicit tax on liquidity transformation.

Interpretation: The LCR reduces short-term debt by making liquidity transformation more costly. But it also forces banks to hold costly liquid assets, so it can crowd out lending.

4.6 Welfare under the LCR

Proposition 3. Welfare under the LCR is lower than the first-best outcome.

Interpretation: The fixed quantity mandate cannot simultaneously set liquidity and lending at first-best levels. This is the core regulatory-design problem.

4.7 Uncertain liquidity demand and procyclicality

Proposition 4. The implicit tax imposed by the LCR is increasing in realized liquidity demand:

$$\frac{\partial \tau^{LCR}}{\partial \xi} > 0. \quad (14)$$

Interpretation: When liquidity demand is high, the liquidity premium rises. Because the LCR's implicit tax depends on the liquidity premium, the regulation becomes more punitive exactly when liquidity transformation is valuable.

4.8 Committed liquidity facility

A committed liquidity facility (CLF) allows banks to pay an upfront fee for a commitment from the central bank. The commitment can count toward the LCR requirement.

Proposition 5. With a correctly designed CLF, the LCR can achieve the first-best equilibrium.

Interpretation: The CLF adds a price-based component to a quantity-based regulation. It can reduce crowding out and limit migration by replacing some required HQLA with a commitment contract.

5 Identification and empirical design

5.1 What identifies the direct effect of LCR exposure

- The key treatment variable is the pre-regulation gap between the required liquidity ratio and the bank's actual liquidity ratio.
- The design compares banks with different gaps before and after LCR implementation.
- Bank fixed effects absorb time-invariant business-model differences.
- Time fixed effects absorb aggregate shocks.
- Controls include leverage, CCAR, SLR, GSIB status, and non-interest income.

Interpretation: The identification assumption is that high-gap and low-gap LCR banks would have followed parallel trends absent the regulation, after controlling for bank characteristics and fixed effects.

5.2 How the paper addresses loan-demand concerns

- The CRA and HMDA data are granular at the bank-MSA level.
- Some regressions include MSA fixed effects and MSA-time fixed effects.
- Mortgage regressions separate conforming and non-conforming lending because conforming loans are often securitized and less balance-sheet intensive.
- The lending decline is stronger in segments that use bank balance-sheet capacity.

Interpretation: These tests aim to isolate bank credit supply from local borrower demand.

5.3 How the paper identifies spillovers

- Non-LCR banks are not directly subject to the LCR.
- The paper measures how exposed each market is to LCR banks before the regulation.
- If LCR banks pull back, non-LCR banks in high-exposure markets should expand lending.
- The paper also tests whether non-LCR banks' liquidity ratios deteriorate in those markets.

Interpretation: The spillover design shows that regulation can move liquidity risk rather than eliminate it.

6 Tables and figures

6.1 Tables 1–3: summary statistics

Read:

- Table 1 compares balance-sheet variables for LCR and non-LCR banks.
- LCR banks are much larger and have lower liquidity ratios than non-LCR banks before the post-LCR period.
- Tables 2 and 3 summarize the CRA small-business and HMDA mortgage samples used in lending tests.

Economic message: The empirical samples show that LCR banks are large, balance-sheet-constrained intermediaries, while non-LCR banks are smaller and more liquid but have weaker access to wholesale funding markets.

Table 1: Summary Statistics of the Bank Balance Sheet Data.

Panel A: LCR banks							
	mean	sd	p5	p25	p50	p75	p95
liquidity coverage ratio	0.822	0.449	0.304	0.482	0.752	1.012	1.688
capital ratio	9.928	2.497	7.110	8.860	9.893	10.910	12.469
og assets	25.821	1.066	24.630	25.138	25.522	26.200	28.148
QLA growth	0.108	0.245	-0.206	-0.010	0.038	0.178	0.748
liquid asset growth	0.032	0.068	-0.090	-0.002	0.017	0.064	0.184
deposits growth	0.049	0.056	-0.036	0.000	0.044	0.082	0.160
non-interest income	0.017	0.010	0.007	0.011	0.016	0.021	0.032
leverage	0.099	0.026	0.065	0.084	0.094	0.107	0.154
leverage (risk-weighted)	0.536	0.433	0.103	0.123	0.188	1.000	1.000
LCAR	0.947	0.223	0.000	1.000	1.000	1.000	1.000
LR	0.077	0.267	0.000	0.000	0.000	0.000	1.000
ISIB	0.192	0.394	0.000	0.000	0.000	0.000	1.000

Panel B: Non-LCR banks							
	mean	sd	p5	p25	p50	p75	p95
liquidity coverage ratio	2.736	2.788	0.328	0.916	1.812	3.477	8.561
capital ratio	9.989	4.081	6.050	8.550	9.650	10.970	14.410
og assets	19.148	1.262	17.321	18.310	19.031	19.834	21.433
QLA growth	0.072	0.309	-0.346	-0.129	0.000	0.207	0.784
liquid asset growth	0.036	0.077	-0.091	-0.004	0.019	0.086	0.184
deposits growth	0.034	0.060	-0.051	0.000	0.018	0.069	0.160
non-interest income	0.007	0.008	0.001	0.003	0.005	0.009	0.019
leverage	0.106	0.033	0.072	0.088	0.100	0.116	0.161
leverage (risk-weighted)	0.861	0.313	0.126	1.000	1.000	1.000	1.000
LCAR	0.000	0.013	0.000	0.000	0.000	0.000	0.000
LR	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ISIB	0.000	0.000	0.000	0.000	0.000	0.000	0.000

te: This table reports the summary statistics of the bank data from Call Reports and FR Y-9C. The unit of observation is a bank-quarter combination. The sample period is from 2011 to 2017.

Table 1: bank balance sheet summary

Table 2: Summary Statistics of the Mortgage Loan Origination Data.

Panel A: LCR banks							
	mean	sd	p5	p25	p50	p75	p95
ummo growth	0.055	0.646	-1.071	-0.356	0.061	0.481	1.127
onforming growth	-0.130	0.687	-1.276	-0.481	-0.111	0.195	1.011
og Assets	19.642	1.217	18.054	18.621	19.367	21.037	21.455
on-interest income	0.017	0.005	0.008	0.013	0.017	0.021	0.025
ortgage share	0.268	0.104	0.108	0.190	0.277	0.329	0.421
l loan share	0.144	0.065	0.061	0.102	0.130	0.190	0.262
leverage	0.091	0.015	0.070	0.083	0.089	0.097	0.115
leverage (risk-weighted)	0.561	0.440	0.098	0.117	1.000	1.000	1.000
LCAR	0.991	0.089	1.000	1.000	1.000	1.000	1.000
ISIB	0.378	0.485	0.000	0.000	0.000	1.000	1.000
LR	0.137	0.308	0.000	0.000	0.000	0.000	1.000

Panel B: Non-LCR banks							
	mean	sd	p5	p25	p50	p75	p95
ummo growth	0.103	0.712	-1.113	-0.374	0.083	0.618	1.277
onforming growth	0.046	0.770	-1.250	-0.355	0.023	0.424	1.515
og Assets	14.426	1.637	12.064	13.168	14.086	15.758	17.239
on-interest income	0.014	0.014	0.002	0.006	0.009	0.015	0.057
ortgage share	0.507	0.152	0.229	0.417	0.516	0.610	0.734
l loan share	0.106	0.073	0.023	0.054	0.089	0.140	0.244
leverage	0.096	0.019	0.072	0.085	0.093	0.104	0.129
leverage (risk-weighted)	0.902	0.273	0.127	1.000	1.000	1.000	1.000
LCAR	0.001	0.014	0.000	0.000	0.000	0.000	0.000
ISIB	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LR	0.000	0.000	0.000	0.000	0.000	0.000	0.000

te: This table reports the summary statistics of the HMDA data. The unit of observation is a bank-MSA-year combination. The sample period is from 2011 to 2017.

to the liability side. Fig. 2(c) shows the main margin of adjustment. Fig. 2(d) shows

Table 3: Summary Statistics of the CRA Loan Origination Data.

Panel A: LCR banks							
	mean	sd	p5	p25	p50	p75	p95
oan growth	-0.026	0.868	-1.524	-0.581	-0.027	0.481	1.600
CR-bank share	0.501	0.200	0.165	0.359	0.506	0.647	0.827
stress-test-bank share	0.645	0.197	0.263	0.515	0.681	0.795	0.917
og Assets	19.380	1.244	17.713	18.335	19.117	20.959	21.425
on-interest income	0.018	0.011	0.007	0.011	0.017	0.020	0.057
ortgage share	0.238	0.136	0.000	0.148	0.250	0.319	0.424
omm. loan share	0.158	0.102	0.003	0.096	0.136	0.198	0.410
leverage	0.096	0.020	0.077	0.084	0.091	0.102	0.136
leverage (risk-weighted)	0.545	0.437	0.101	0.119	0.201	1.000	1.000
LCAR	0.946	0.226	0.000	1.000	1.000	1.000	1.000
ISIB	0.264	0.441	0.000	0.000	0.000	1.000	1.000
LR	0.150	0.357	0.000	0.000	0.000	0.000	1.000

Panel B: Non-LCR banks							
	mean	sd	p5	p25	p50	p75	p95
oan growth	0.060	0.994	-1.669	-0.612	0.002	0.750	1.778
CR-bank share	0.503	0.195	0.167	0.359	0.517	0.649	0.799
stress-test-bank share	0.653	0.187	0.279	0.535	0.696	0.795	0.905
og Assets	15.635	1.205	13.777	14.571	15.684	16.720	17.448
on-interest income	0.014	0.012	0.002	0.006	0.011	0.017	0.052
ortgage share	0.386	0.180	0.001	0.271	0.415	0.518	0.635
omm. loan share	0.185	0.166	0.031	0.078	0.124	0.235	0.580
leverage	0.101	0.027	0.073	0.086	0.094	0.110	0.158
leverage (risk-weighted)	0.837	0.337	0.111	1.000	1.000	1.000	1.000
LCAR	0.016	0.124	0.000	0.000	0.000	0.000	0.000
ISIB	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LR	0.000	0.000	0.000	0.000	0.000	0.000	0.000

te: This table reports the summary statistics of the CRA data. The unit of observation is a bank-MSA-year combination. The sample period is from 2011 to 2017.

Table 2: CRA loan summary

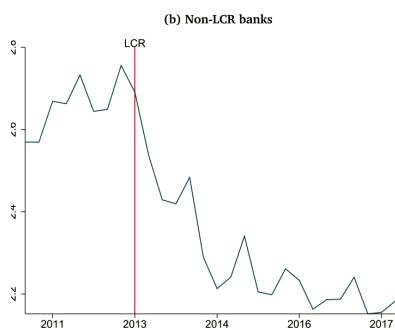
6.2 Figures 1–3: aggregate trends and event-study evidence

Read:

- Figure 1 shows that LCR banks' liquidity ratios improve after 2013 while non-LCR banks' ratios decline.
- Figure 2 shows that LCR banks increase HQLA holdings but grow illiquid assets more slowly than non-LCR banks.
- Deposits grow similarly, but non-deposit funding diverges.

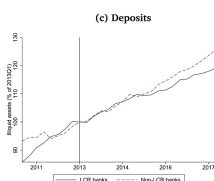
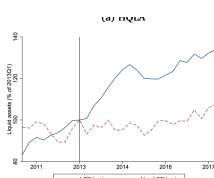
- Figure 3 shows dynamic treatment effects: HQLA holdings rise for high-gap LCR banks after implementation.

Economic message: The aggregate and dynamic evidence shows the first direct effect of the LCR: regulated banks substantially improve measured liquidity.

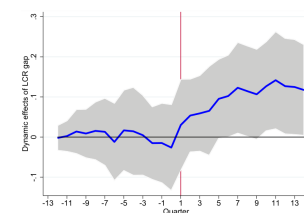
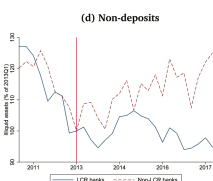
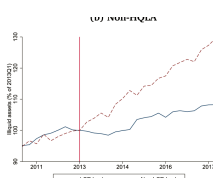


1. Liquidity Coverage Ratio of U.S. Banks. This figure plots the liquidity

(a) Figure 1: LCR ratios.



(b) Figure 2: balance-sheet components.



(c) Figure 3: dynamic HQLA effects.

6.3 Tables 4–5: HQLA and illiquid asset holdings

Read:

- Table 4 shows that more-exposed LCR banks increase HQLA holdings after the LCR.
- The coefficient on $Post \times Liquidity\ ratio\ gap$ is positive and significant across specifications.
- Table 5 shows that more-exposed banks reduce illiquid assets after the LCR.
- The negative illiquid-asset coefficient is consistent with crowding out.

Economic message: The LCR makes banks safer in a narrow liquidity sense, but this comes at the cost of reallocating balance-sheet capacity away from illiquid assets.

Table 4
LCR Banks' HQLAs Holdings.

	(1) HQLA	(2) HQLA	(3) HQLA
Post * Liquidity ratio gap	0.307*** [0.107]	0.372*** [0.131]	0.369*** [0.120]
Post * Non-interest income	0.163 [2.778]	-1.544 [2.903]	-1.673 [3.013]
Post * Leverage	-2.581** [1.239]	-3.089** [1.508]	-3.081* [1.571]
Post * Leverage (risk-weighted)	0.254* [0.137]	0.110 [0.114]	0.132 [0.122]
Post * CCAR	0.102 [0.159]	0.053 [0.170]	0.048 [0.170]
Post * GSIB	-0.142* [0.082]	-0.046 [0.095]	-0.050 [0.118]
Post * SLR	-0.017 [0.030]	0.042 [0.041]	0.062 [0.085]
Control	Yes	Yes	Yes
Bank F.E.	Yes	Yes	Yes
Time F.E.	No	Yes	No
Category-Time F.E.	No	No	Yes
Observations	832	832	832
Adj. R-squared	0.275	0.291	0.275

Note: This table shows the effect of liquidity regulation on LCR banks' asset holdings. The sample includes banks with assets above \$50 billion from 2011Q1 to 2017Q4. The dependent variable is the growth rates of high-quality liquid assets. *Post* is a dummy variable that equals one if the time is after 2013Q1. *Liquidity ratio gap* is the difference between the required liquidity ratio and the pre-regulation liquidity ratio. *Non-interest income* is the non-interest income normalized by assets. *Leverage* is the leverage ratio with unweighted assets. *Leverage (risk-weighted)* is the leverage ratio with risk-weighted assets. *CCAR* is a dummy variable that equals one if a bank is subject to CCAR stress tests, and zero otherwise. *GSIB* is a dummy variable that equals one if a bank is a global systemically important bank, and zero otherwise. *SLR* is a dummy variable that equals one if a bank is subject to the supplementary leverage ratio requirement, and zero otherwise. The control variables also include log assets, the share

Table 4: HQLA holdings

Table 5
LCR Banks' Illiquid Assets Holdings.

	(1) Illiquid assets	(2) Illiquid assets	(3) Illiquid assets
Post * Liquidity ratio gap	-0.084*** [0.022]	-0.055* [0.027]	-0.061* [0.030]
Post * Non-interest income	-0.256 [0.717]	-0.726 [0.643]	-0.593 [0.619]
Post * Leverage	-0.451* [0.221]	-0.292 [0.308]	-0.333 [0.315]
Post * Leverage (risk-weighted)	0.038** [0.015]	0.013 [0.018]	0.012 [0.020]
Post * CCAR	-0.058 [0.048]	-0.050 [0.041]	-0.043 [0.043]
Post * GSIB	-0.001 [0.019]	0.026 [0.033]	0.031 [0.033]
Post * SLR	-0.016** [0.007]	0.005 [0.009]	-0.010 [0.012]
Control	Yes	Yes	Yes
Bank F.E.	Yes	Yes	Yes
Time F.E.	No	Yes	No
Category-Time F.E.	No	No	Yes
Observations	832	832	832
Adj. R-squared	0.339	0.417	0.403

Note: This table shows the effect of liquidity regulation on LCR banks' asset holdings. The sample includes banks with assets above \$50 billion from 2011Q1 to 2017Q4. The dependent variable is the growth rates of illiquid assets. *Post* is a dummy variable that equals one if the time is after 2013Q1. *Liquidity ratio gap* is the difference between the required liquidity ratio and the pre-regulation liquidity ratio. *Non-interest income* is the non-interest income normalized by assets. *Leverage* is the leverage ratio with unweighted assets. *Leverage (risk-weighted)* is the leverage ratio with risk-weighted assets. *CCAR* is a dummy variable that equals one if a bank is subject to CCAR stress tests, and zero otherwise. *GSIB* is a dummy variable that equals one if a bank is a global systemically important bank, and zero otherwise. *SLR* is a dummy variable that equals one if a bank is subject to the supplementary leverage ratio requirement, and zero otherwise. The control variables also include log assets, the share of real estate loans, and the share of commercial and industrial loans. *Category* indicates whether a bank is subject to full LCR or modified LCR. Standard errors shown in parentheses are clustered at the bank level.

Table 5: illiquid assets

6.4 Tables 6–7: lending effects on LCR banks

Read:

- Table 6 shows lower small-business lending growth by more-exposed LCR banks.
- A 10% higher liquidity-ratio gap is associated with roughly 10% lower annual loan growth after the LCR.
- Table 7 shows that the lending reduction is stronger for non-conforming mortgages than for conforming mortgages.
- This pattern fits the balance-sheet crowding-out interpretation because non-conforming mortgages are less easily securitized.

Economic message: Liquidity regulation has real credit-supply consequences. It reduces lending in bank-balance-sheet-intensive markets.

Table 6
Small Business Lending of LCR Banks.

	(1) Loans	(2) Loans	(3) Loans
Post * Liquidity ratio gap	-1.001*** [0.320]	-1.008*** [0.317]	-1.056*** [0.319]
Post * Non-interest income	-2.588 [8.181]	-1.999 [7.971]	-3.656 [8.180]
Post * Leverage	-1.422 [3.483]	-1.313 [3.495]	-1.614 [3.885]
Post * Leverage (risk-weighted)	0.055 [0.265]	0.070 [0.256]	0.109 [0.256]
Post * CCAR	0.119 [0.333]	0.167 [0.345]	0.123 [0.379]
Post * GSIB	-0.655** [0.308]	-0.657** [0.312]	-0.746** [0.322]
Post * SLR	0.107 [0.104]	0.124 [0.101]	0.141 [0.109]
Control	Yes	Yes	Yes
Bank F.E.	Yes	Yes	No
MSA F.E.	Yes	No	No
Time F.E.	Yes	No	No
MSA-Time F.E.	No	Yes	Yes
MSA-Bank F.E.	No	No	Yes
Observations	34,051	34,051	33,145
Adj. R-squared	0.075	0.079	0.036

Note: This table shows the effect of liquidity regulation on LCR banks' CRA lending. The sample includes banks with assets above \$50 billion from 2011Q1 to 2017Q4. The dependent variable is the growth rate of small business loans. *Post* is a dummy variable that equals one if the time is after 2013Q1. *Liquidity ratio gap* is the difference between the required liquidity ratio and the pre-regulation liquidity ratio. *Non-interest income* is the non-interest income normalized by assets. *Leverage* is the leverage ratio with unweighted assets. *Leverage (risk-weighted)* is the leverage ratio with risk-weighted assets. *CCAR* is a dummy variable that equals one if a bank is subject to CCAR stress tests, and zero otherwise. *GSIB* is a dummy variable that equals one if a bank is a global systemically important bank, and zero otherwise. *SLR* is a dummy variable that equals one if a bank is subject to the supplementary leverage ratio requirement, and zero otherwise. The control variables also include

Table 6: small-business lending

Table 7

Mortgage Lending of LCR Banks.

	(1) Non-conforming	(2) Non-conforming	(3) Conforming	(4) Conforming
Post * Liquidity ratio gap	-0.313** [0.136]	-0.474** [0.207]	0.095 [0.109]	0.305 [0.216]
Post * Non-interest income		6.671 [8.552]		-4.777 [9.283]
Post * Leverage		-2.324 [3.254]		-3.005 [4.159]
Post * Leverage (risk-weighted)		0.180 [0.303]		-0.714* [0.392]
Post * CCAR		-0.790* [0.435]		-0.559 [0.490]
Post * GSIB		-0.054 [0.090]		0.089 [0.145]
Post * SLR		-0.199* [0.110]		-0.013 [0.117]
Control	No	Yes	No	Yes
Time F.E.	Yes	Yes	Yes	Yes
Bank F.E.	Yes	Yes	Yes	Yes
Observations	16,790	16,790	16,356	16,356
Adj. R-squared	-0.074	-0.064	0.114	0.127

This table shows the effect of liquidity regulation on LCR banks' mortgage lending. The sample includes banks with assets above \$50 billion from 2011Q1 to 2017Q4. The dependent variable is the growth rate of non-conforming mortgages (columns 1–2) and conforming mortgages (columns 3–4). *Post* is a dummy variable that equals one if the time is after 2013Q1. *Liquidity ratio gap* is the difference between the required liquidity ratio and the pre-regulation liquidity ratio. *Non-interest income* is the non-interest income normalized by assets. *Leverage* is the leverage ratio with unweighted assets. *Leverage (risk-weighted)* is the leverage ratio with risk-weighted assets. *CCAR* is a dummy variable that equals one if a bank is subject to CCAR stress tests, and zero otherwise. *GSIB* is a dummy variable that equals one if a bank is a global systemically important bank, and zero otherwise. *SLR* is a dummy variable that equals one if a bank is subject to the supplementary leverage ratio requirement, and zero otherwise. The control variables also include assets, the share of real estate loans, and the share of commercial and industrial loans. Standard errors in parentheses are clustered at both the MSA and the bank level. *** **, and *

Table 7: mortgage lending

6.5 Table 8: spillover to non-LCR banks

Read:

- Panel A shows that non-LCR banks increase lending in markets where LCR-bank shares are higher.
- Panel B shows that non-LCR banks' liquidity ratios deteriorate in those same markets.
- The lending coefficient is positive, while the liquidity-ratio coefficient is negative.

Economic message: The LCR shifts some liquidity transformation from regulated banks to unregulated banks. This is risk migration, not pure risk reduction.

Table 8
The Spillover Liquidity Regulation to Non-LCR Banks.

Panel A: Lending			
	(1) Loan growth	(2) Loan growth	(3) Loan growth
Post * LCR-bank share	0.293*** [0.061]	0.306*** [0.061]	0.334*** [0.087]
Control	No	No	Yes
MSA F.E.	No	Yes	Yes
Bank-Time F.E.	Yes	Yes	Yes
Observations	49,193	49,193	49,193
Adj. R-squared	0.098	0.097	0.097
Panel B: Liquidity ratio			
	(1) Liquidity ratio	(2) Liquidity ratio	(3) Liquidity ratio
Post * LCR-bank share	-0.472** [0.206]	-0.386* [0.197]	-0.388* [0.201]
Control	No	No	Yes
Bank F.E.	No	Yes	Yes
Time F.E.	Yes	Yes	Yes
Observations	15,442	15,442	15,442
Adj. R-squared	0.801	0.815	0.816

Note: Panel A shows the spillover effect of liquidity regulation on the loan growth and liquidity ratio of non-LCR banks. The sample includes banks in the CRA data with assets below \$50 billion from 2011Q1 to 2017Q4. The dependent variable, *Loan growth*, is the growth rate in small business loan origination by bank i in MSA m at time t . *Post* is a dummy variable that equals one if the time is after 2013Q1. *LCR-bank share* is defined as the share of loans originated by LCR banks in an MSA in 2012. Bank-time fixed effects and MSA fixed effects are added to the regressions, as shown in the table. Control variables include the local stress-test-bank share, local unemployment rates, and their interactions with *Post* dummy. Standard errors shown in parentheses are clustered at both the MSA level and the bank level. Panel B shows the spillover effect of liquidity regulation on the liquidity coverage ratio of non-LCR banks. The sample includes banks with assets below \$50 billion from 2011Q1 to 2017Q4. The dependent variable *Liquidity ratio* is the liquidity coverage ratio of bank i at time t . *Post* is a dummy variable that equals one if the time is after 2013Q1. *LCR-bank share* is defined as the average share of LCR banks in the MSAs in which a non-LCR bank operates. Bank fixed and time fixed effects are added to the regressions, as shown in the table. Control variables include the local stress-test-bank share and the interaction with *Post* dummy. Standard errors shown in parentheses are clustered at the bank level. ***, **, and * represent 1%, 5%, and 10% significance levels, respectively.

6.6 Figures 4–6: model equilibria and regulatory design

Read:

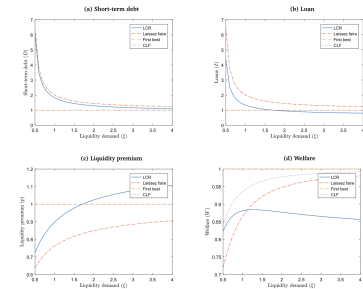
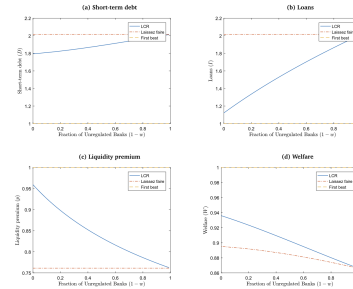
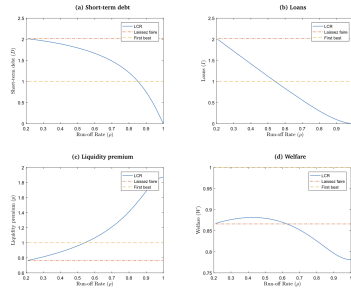
- Figure 4 varies the run-off rate. Tighter LCR requirements reduce short-term debt and lending, raise liquidity premia, and can reduce welfare if too strict.
- Figure 5 varies the fraction of unregulated banks. More unregulated banks soften the effect on regulated banks but increase migration.
- Figure 6 varies liquidity demand. With a fixed LCR, high liquidity demand raises liquidity premia and can reduce welfare. The CLF improves outcomes by reducing the quantity-mandate distortion.

Economic message: The model figures show why the LCR can improve on laissez-faire while still being suboptimal. The CLF helps because it introduces a price-based policy instrument.

6.7 Tables 9–10: calibration and counterfactuals

Read:

- Table 9 reports calibrated parameters and targeted moments for the quantitative policy experiment.
- Table 10 compares laissez-faire, first-best, LCR, stricter LCR, and LCR plus CLF.



- The LCR improves welfare relative to laissez-faire but leaves substantial distortion.
- A stricter LCR worsens welfare because it intensifies crowding out and migration.
- LCR plus CLF closes more of the welfare gap by reducing migration and lending distortions.

Economic message: The calibration makes the design lesson quantitative: simply tightening the LCR is not necessarily better, but adding a CLF can improve welfare.

Parameter	Value	Targeted moments	Value
u_n : mean of log random withdrawal	-0.019	mean of random withdrawal	0.000
τ_n : volatility of log random withdrawal	0.194	expected outflows in distress	0.300
u_m : mean of log demand for liquidity	0.015	mean of liquidity premium	0.007
τ_m : volatility of log demand for liquidity	0.005	volatility of liquidity premium	0.005
r : issuance cost of short-term debt	0.006	ratio of govt to bank liquidity	1.250
ϵ : elasticity of demand for liquidity	58.000	bank liquidity supply	0.800
β : liquidation cost	0.115	pre-LCR liquidity share	0.270
γ : fire-sale externality	0.259	post-LCR liquidity share	0.320
g : govt liquidity	1.000	normalized to one	1.000
w : share of regulated banks	0.800	LCR bank share	0.800

Note: This table shows the calibrated parameters for the policy experiments. Columns (1) and (2) show the parameter value. Columns (3) and (4) show the targeted moments in the data.

which is linked to the borrowing as a normal continuous distribution for the withdrawal. In

7 Short summary

- The LCR improves measured liquidity at regulated banks.
- More-exposed LCR banks increase HQLA holdings after the rule.
- More-exposed LCR banks reduce illiquid assets.
- The mandated buffer crowds out small-business lending and balance-sheet-intensive mortgage lending.
- Non-LCR banks expand lending in markets with high LCR-bank presence.
- Non-LCR banks' liquidity ratios deteriorate in those markets.
- The model explains these facts using liquidity premia, fire-sale externalities, and heterogeneous regulated/unregulated intermediaries.
- The LCR acts as an implicit tax on liquidity transformation and a costly liquidity buffer at the same time.
- This dual role prevents the LCR from achieving first-best lending and liquidity.
- A fixed quantity rule becomes procyclical when liquidity demand is uncertain.

- A committed liquidity facility improves the design by adding a price-based instrument.

8 Conclusion

8.1 Core conclusion

Sundaresan and Xiao show that liquidity regulation has both benefits and costs. The LCR reduces liquidity risk at regulated banks, but it also crowds out lending and shifts liquidity transformation toward unregulated banks. These findings motivate a model in which the LCR improves on laissez-faire but cannot achieve first best because the liquidity buffer simultaneously acts as an implicit tax and a costly asset requirement.

8.2 Economic intuition

Liquidity transformation creates a fire-sale externality, so laissez-faire banks issue too much short-term debt. Regulation can improve outcomes by making liquidity transformation more costly. However, a fixed HQLA requirement also occupies balance-sheet space and raises the liquidity premium. This distorts lending, encourages migration to non-LCR banks, and becomes more punitive in high-liquidity-demand states.

8.3 Takeaway

The paper’s main policy message is that the design of liquidity regulation matters. A stricter LCR is not always better. A committed liquidity facility can improve the current quantity-based rule by replacing part of the mandated HQLA buffer with a commitment fee, reducing crowding out and risk migration while preserving liquidity insurance.